

Handling Ambiguous SSA Cases Completely

Answer Key

Precalculus Trigonometry and Vectors

Quick Answers

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|--------------------------------|--|
| 1. 5.00, 56.44 | 6. 90, 6.00, 10.39 |
| 2. 58.99, 7.71, 1.10 | 7. 36.06, 50.46 |
| 3. 63.15, 116.85, 15.53, 7.40 | 8. 9.00, 64.16, 115.84, 19.95, 11.23, 1.00 |
| 4. 31.35, 196.65, 13.22 | 9. 64.56, 115.44, 27.29, 10.11 |
| 5. 60.62, 119.38, 45.81, 21.28 | 10. 61.10, 12.26 |
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Curriculum

Level: Precalculus Trigonometry and Vectors

HSG-SRT.D.10, HSG-SRT.D.11 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 36 tagged checks.

Common wrong answers

1. *If they got 3.00: computed a $\sin A$ instead of $b \sin A$, swinging the wrong side down to the base line*
 3. *If they got 63.15: stopped at the acute angle and never reported the second triangle*
 5. *If they got 60.62: reported only the acute angle and never checked whether the supplement also fits*
 8. *If they got 15.59: used the cosine instead of the sine when computing the height*
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1. Case test: $A = 30^\circ$, $b = 10$, $a = 6$.

Step 1. Part (a): the height is the perpendicular from C to the base line, $h = b \sin A = 10 \sin 30^\circ$. $h = 5.00$

Step 2. Part (b): $h = 5 < a = 6 < b = 10$, so the swinging side is long enough to reach the base line but short enough to reach it twice — two triangles.

Step 3. The law of sines gives $\sin B = \frac{b \sin A}{a} = \frac{5}{6} = 0.8333$, so $B = \arcsin 0.8333$.

$B = 56.44^\circ$

Why h comes first: it is the only number that decides the case, and it costs one multiplication. Students who skip it end up discovering the second triangle by luck.

2. Two sightings: $A = 40^\circ$, $b = 12$, first $a = 9$.

Step 1. Part (a): $h = 12 \sin 40^\circ$.

$$h = 7.71$$

Step 2. Since $7.71 < 9 < 12$, the measurement is ambiguous: two triangles. The acute solution comes from $\sin B = \frac{12 \sin 40^\circ}{9} = 0.8571$.

$$B = 58.99^\circ$$

Step 3. Part (b): with $a = 7$ the same rule gives $\sin B = \frac{12 \sin 40^\circ}{7}$.

$$\sin B = 1.10$$

Step 4. No angle has a sine above 1, so no triangle exists — which matches the height test, since $a = 7$ is shorter than $h = 7.71$ and the swinging side cannot reach the base line at all. The second sighting must be a measurement error.

Grading note: accept either justification in part (b) — the impossible sine or the comparison with h . They are the same fact stated two ways. Note how narrow the miss is: the side is short by less than a unit, so this case cannot be settled by looking at the numbers, only by computing h .

3. Both triangles: $A = 35^\circ$, $b = 14$, $a = 9$.

Step 1. $h = 14 \sin 35^\circ = 8.03$, and $8.03 < 9 < 14$, so expect two triangles. From $\sin B = \frac{14 \sin 35^\circ}{9} = 0.8922$:

$$B_1 = 63.15^\circ$$

Step 2. The supplement has the same sine: $B_2 = 180^\circ - 63.15^\circ$. Test it: $A + B_2 = 35^\circ + 116.85^\circ = 151.85^\circ < 180^\circ$, so a third angle of 28.15° remains and the second triangle is real.

$$B_2 = 116.85^\circ$$

Step 3. Triangle 1: $C_1 = 180^\circ - 35^\circ - 63.15^\circ = 81.85^\circ$, and $c_1 = \frac{a \sin C_1}{\sin A} = \frac{9 \sin 81.85^\circ}{\sin 35^\circ}$.

$$c_1 = 15.53$$

Step 4. Triangle 2: $C_2 = 180^\circ - 35^\circ - 116.85^\circ = 28.15^\circ$, and $c_2 = \frac{9 \sin 28.15^\circ}{\sin 35^\circ}$.

$$c_2 = 7.40$$

The check that catches everything: in each triangle the largest side must face the largest angle. Triangle 1 has C_1 largest and $c_1 = 15.53$ largest; Triangle 2 has B_2 largest and $b = 14$ largest. Both pass.

4. Only one: $A = 48^\circ$, $b = 7$, $a = 10$.

Step 1. Part (a): $\sin B = \frac{7 \sin 48^\circ}{10} = 0.5202$.

$$B = 31.35^\circ$$

Step 2. Part (b): the obtuse candidate is $180^\circ - 31.35^\circ = 148.65^\circ$, and adding the given angle gives $48^\circ + 148.65^\circ$.

$$A + B_2 = 196.65^\circ$$

Step 3. That exceeds 180° , so no triangle can contain both angles and the obtuse candidate is rejected. This is what $a \geq b$ always forces: the side opposite the larger angle is longer, so $a > b$ makes $A > B$ and B cannot be obtuse.

Step 4. Part (c): $C = 180^\circ - 48^\circ - 31.35^\circ = 100.65^\circ$ and $c = \frac{10 \sin 100.65^\circ}{\sin 48^\circ}$.

$$c = 13.22$$

Worth saying aloud: “one triangle” here is not luck — it is guaranteed the moment $a \geq b$, before any arithmetic.

5. Search and rescue: $A = 33^\circ$, $b = 40$, $a = 25$.

Step 1. $h = 40 \sin 33^\circ = 21.79$, and $21.79 < 25 < 40$: the bearings allow two positions. From $\sin B = \frac{40 \sin 33^\circ}{25} = 0.8714$:

$$B_1 = 60.62^\circ$$

Step 2. The supplement passes the test ($33^\circ + 119.38^\circ = 152.38^\circ < 180^\circ$), so it is a genuine second position.

$$B_2 = 119.38^\circ$$

Step 3. Far position: $C_1 = 180^\circ - 33^\circ - 60.62^\circ = 86.38^\circ$, so $c_1 = \frac{25 \sin 86.38^\circ}{\sin 33^\circ}$.

$$c_1 = 45.81 \text{ km}$$

Step 4. Near position: $C_2 = 180^\circ - 33^\circ - 119.38^\circ = 27.62^\circ$, so $c_2 = \frac{25 \sin 27.62^\circ}{\sin 33^\circ}$.

$$c_2 = 21.28 \text{ km}$$

Why this matters operationally: the two answers are 45.81 km and 21.28 km along the same shoreline — roughly 24.5 km apart. Reporting only the first is not a rounding error; it is a search in the wrong place, and it is exactly what the law of sines will hand you if you never test the supplement.

6. The borderline: $A = 30^\circ$, $b = 12$, $a = 6$.

Step 1. Part (a): $h = 12 \sin 30^\circ$.

$$h = 6.00$$

Step 2. Here $a = h$ exactly. The swinging side just reaches the base line, meeting it at the foot of the perpendicular, so the angle at B is a right angle: $\sin B = \frac{12 \sin 30^\circ}{6} = 1$.

$$B = 90^\circ$$

Step 3. Part (b): the supplement of 90° is 90° again — there is no second angle to find. The two solutions of the ambiguous case have merged into one.

Step 4. Part (c): $C = 180^\circ - 30^\circ - 90^\circ = 60^\circ$, so $c = \frac{6 \sin 60^\circ}{\sin 30^\circ}$.

$$c = 10.39$$

The boundary case is the useful one: it is the exact moment the two triangles of the ambiguous case collapse into each other, which is why $a = h$ separates “none” from “two”. Compare problem 1: same swinging side, same angle, and only b changed — lengthening b raised h until it caught up with a , turning two triangles into one.

7. Surveying: $A = 62^\circ$, $b = 30$, $a = 45$.

Step 1. Part (a): $a = 45$ is longer than $b = 30$, so the angle opposite a is the larger of the two and B must be acute. No supplement test is needed — one triangle, guaranteed. Then $\sin B = \frac{30 \sin 62^\circ}{45} = 0.5886$.

$$B = 36.06^\circ$$

Step 2. Part (b): $C = 180^\circ - 62^\circ - 36.06^\circ = 81.94^\circ$, so $c = \frac{45 \sin 81.94^\circ}{\sin 62^\circ}$.

$$c = 50.46 \text{ m}$$

Check: the largest angle here is $C = 81.94^\circ$ and the largest side is $c = 50.46$ m. Consistent.

8. How much slack: $A = 30^\circ$, $b = 18$, $a = 10$.

Step 1. Part (a): $h = 18 \sin 30^\circ$. At $A = 30^\circ$ the height is exactly half of b , so this measurement is ambiguous precisely when $9 < a < 18$.

$$h = 9.00$$

Step 2. $9 < 10 < 18$, so two triangles. From $\sin B = \frac{18 \sin 30^\circ}{10} = 0.9000$:

$$B_1 = 64.16^\circ$$

Step 3. The supplement passes ($30^\circ + 115.84^\circ = 145.84^\circ < 180^\circ$).

$$B_2 = 115.84^\circ$$

Step 4. Part (b), Triangle 1: $C_1 = 85.84^\circ$ and $c_1 = \frac{10 \sin 85.84^\circ}{\sin 30^\circ}$.

$$c_1 = 19.95$$

Step 5. Triangle 2: $C_2 = 34.16^\circ$ and $c_2 = \frac{10 \sin 34.16^\circ}{\sin 30^\circ}$.

$$c_2 = 11.23$$

Step 6. Part (c): the slack is $a - h = 10 - 9$.

$$a - h = 1.00$$

Step 7. Part (c), second half — open response, read the reasoning. Shrink a by that much and $a = h$: the swinging side is then exactly as long as the perpendicular, so it meets the base line once, at the foot of the perpendicular. One triangle remains and it is right-angled at B — the situation of problem 6.

What this problem is really about: this is the tightest ambiguous measurement on the sheet, and one unit of slack is inside the error of many real instruments. A student who reports one triangle here is not making an arithmetic mistake; they are reporting a measurement they never tested. *Accept* any answer that identifies exactly one surviving triangle and names the right angle at B (or, equivalently, $\sin B = 1$).

9. Two possible positions: $A = 44^\circ$, $b = 26$, $a = 20$.

Step 1. $h = 26 \sin 44^\circ = 18.06 < 20 < 26$, so both positions are geometrically valid. From $\sin B = \frac{26 \sin 44^\circ}{20} = 0.9031$:

$$B_1 = 64.56^\circ$$

Step 2. The supplement passes ($44^\circ + 115.44^\circ = 159.44^\circ$).

$$B_2 = 115.44^\circ$$

Step 3. With $C_1 = 71.44^\circ$: $c_1 = \frac{20 \sin 71.44^\circ}{\sin 44^\circ}$.

$$c_1 = 27.29$$

Step 4. With $C_2 = 20.56^\circ$: $c_2 = \frac{20 \sin 20.56^\circ}{\sin 44^\circ}$.

$$c_2 = 10.11$$

Step 5. Part (b) — open response, read the reasoning. The obtuse solution $B_2 = 115.44^\circ$ puts the marker between the control point and the drone (the near apex, $c_2 = 10.11$), so “the drone is beyond the marker” rules it out and leaves the acute triangle with $c_1 = 27.29$.

What a full answer names: any single extra measurement that separates the two — the distance c itself, the angle at C , or whether the angle at B is acute or obtuse. Accept any of these; the point is that ONE more observation is enough, and that without it both answers must be reported.

10. Challenge — the whole case rule: $A = 50^\circ$, $b = 16$.

Step 1. Part (a): with $a = 14$, $\sin B = \frac{16 \sin 50^\circ}{14} = 0.8755$.

$$B_1 = 61.10^\circ$$

Step 2. The supplement 118.90° also passes ($50^\circ + 118.90^\circ = 168.90^\circ < 180^\circ$), so $a = 14$ gives two triangles.

Step 3. Part (b): $h = 16 \sin 50^\circ$.

$$h = 12.26$$

Step 4. Part (c) — open response, read the reasoning. The complete rule for $A = 50^\circ$, $b = 16$:

- $a < 12.26$: no triangle. The swinging side is shorter than the perpendicular and never reaches the base line.
- $a = 12.26$: exactly one, the right triangle where the side meets the base line at the foot of the perpendicular.
- $12.26 < a < 16$: two triangles. The side is long enough to cross the base line on both sides of the perpendicular.
- $a \geq 16$: exactly one. The side now reaches past the vertex A , so only the far intersection lies on the correct side of A ; equivalently $a \geq b$ forces B acute.

Accept any statement of these four bands with the two boundary values 12.26 and 16 identified, justified either by the swinging side or by the sine test ($\sin B > 1$, $= 1$, or two admissible angles).